

A PROJECT REPORT

ON

**DESIGN AND IMPLEMENTATION OF A SECURE AND SCALABLE
NETWORK INFRASTRUCTURE FOR A SHOPPING MALL**

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CERTIFICATE

This is to certify that the project work entitled " DESIGN AND IMPLEMENTATION OF A SECURE AND SCALABLE NETWORK INFRASTRUCTURE FOR A SHOPPING MALL" using CISCO PACKET TRACER is a bonafide work of Mr. Bijoy Simon. I have successfully completed the project work at NETWORKZ SYSTEMS during the period from 07-07-2025 TO 1-09-2025 towards the partial Fulfillment of the requirement for the award of Network training program

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ACKNOWLEDGEMENT

At the very outset, I would like to give the first honors to God, who gave the wisdom and knowledge to complete the project. With profound sense of gratitude, I wish to express my sincere thanks to Mr. **SAJANLAL S.** I am thankful to all other staff members necessary help to enable me to undertake my project with keen of **Networkz Systems, Kollam** for their support and guidance.

Finally, I thank my parents and all my friends and well-wishers who help in the successful completion of this project. Last, but not least I thank the almighty without whose blessings I would not have succeeded in my endeavours.

BIJOY SIMON

DECLARATION

I hereby declare that this project work entitled " **DESIGN AND IMPLEMENTATION OF A SECURE AND SCALABLE NETWORK INFRASTRUCTURE FOR A SHOPPING MALL** " submitted to "NETWORKZ SYSTEMS" is a record of original work done by me under the Guidance of Mr. **SAJANLAL S**, Networkz Systems have not formed the basis for the award of a training program.

Abstract

This project simulates the design and implementation of a secure, scalable enterprise-level network for a shopping mall using Cisco Packet Tracer. The network follows a three-tier hierarchical architecture with a core router, two Layer 3 distribution switches, and eight access layer switches. Multiple VLANs segment the network into zones for retail stores, a public food court, CCTV surveillance, administrative management, internal service servers, and Internet of Things (IoT) devices. Additionally, a dedicated server services VLAN is implemented on a separate access switch that is directly connected to one of the distribution switches.

One access switch supports VLAN 80, which connects IoT systems such as fire alarms, sprinklers, air conditioning units, and detectors. These devices are monitored and managed remotely from authorized control systems. A wireless access point is also deployed in VLAN 80 to support wireless IoT devices that require Wi-Fi connectivity.

The core router connects to a simulated ISP via an additional router, which includes a loopback interface configured with IP address 8.8.8.8 to represent internet access. A switch connected to the ISP router hosts external services such as HTTP, FTP, and Syslog servers for mall systems to access. The core router provides NAT to enable internal devices to reach external resources and runs OSPF for dynamic routing with the distribution switches. A passive OSPF interface is configured between the distribution switches to enhance security and prevent unnecessary neighbor relationships. Inter-VLAN routing is handled through SVIs on the distribution switches, which are also configured with Hot Standby Router Protocol (HSRP) to ensure gateway redundancy and seamless failover.

Each distribution switch also functions as a DHCP server, dynamically assigning IP addresses to the VLANs under its control. DHCP pools are carefully split to avoid IP conflicts and ensure high availability in case of a switch failure. Additionally, a DHCP server is added to VLAN 80 to dynamically assign IP addresses to IoT devices such as fire alarms, sprinklers, air conditioning units, and detectors. At the access layer, each switch is mapped to a dedicated VLAN. A wireless access point is deployed in the food court VLAN (VLAN 60) to provide Wi-Fi access for customers and their personal devices.

Network security is enforced using Access Control Lists (ACLs) that isolate communication between retail, food court, CCTV, IoT, and server VLANs, while allowing full access from the administrative VLAN. The server services switch, hosting VLANs for HTTP, FTP, and TFTP servers, is securely integrated and accessible only to authorized segments. SSH is enabled for secure remote management, and port security is implemented across all access switches to prevent unauthorized device connections based on MAC address restrictions. To ensure loop-free Layer 2 connectivity across redundant links, Spanning Tree Protocol (STP) is enabled by default, with PortFast configured on access ports to accelerate convergence.

A centralized syslog server, located beyond the core router, collects logs from internal devices for real-time monitoring and auditing. Overall, this project demonstrates a well-structured, fault-tolerant, and secure network infrastructure suitable for a commercial mall—featuring effective segmentation, dynamic IP management, wireless access, and redundancy.

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NETWORK REQUIREMENTS

- The network must provide secure and reliable connectivity for the admin office, retail store, CCTV surveillance system, and food counter Wi-Fi access.
- **DHCP** service
- Interdepartmental Connectivity (Inter-VLAN)
- **OSPF** dynamic routing protocol ensures optimal path selection between core and distribution layers.
- **HSRP** provides gateway redundancy for fault tolerance.
- **Access Control Lists (ACLs)** enforce communication policies:
 - Departments cannots communicate with each other, except admin can sent traffic to all deparments.
- **Syslog** Server, placed beyond the core router, collects device logs for centralized monitoring.
- All VLANs can access external servers and the internet via **NAT** on the core router, with simulated ISP integration.

INTRODUCTION

In today's digitally connected environments, modern commercial infrastructures such as shopping malls require robust, secure, and scalable networks to support diverse operational needs. These needs range from administrative communications and retail operations to security surveillance, customer Wi-Fi, and the growing integration of Internet of Things (IoT) devices. Designing such a network involves not only connecting devices but also ensuring secure data flow, service availability, and resilience against failures.

This project presents the design and simulation of an enterprise-level network for a multi-zone shopping mall using Cisco Packet Tracer. The network follows a three-tier hierarchical model comprising core, distribution, and access layers. It integrates multiple VLANs to segment traffic logically for different functional areas such as retail, administration, food courts, CCTV systems, servers, and IoT equipment.

To ensure reliability and efficiency, the design incorporates key enterprise networking features including dynamic routing via OSPF, gateway redundancy with HSRP, DHCP-based IP address management, secure remote access, and access control policies. Wireless access points are also deployed for both public and internal device connectivity.

Network Design Overview

1. Network Design Models

This network is based on the three-tier hierarchical architecture, a widely adopted model that separates the network into core, distribution, and access layers.

- The core layer provides fast and reliable backbone connectivity and links to the ISP.
- The distribution layer aggregates traffic from access switches and performs inter-VLAN routing and policy enforcement.
- The access layer connects end-user devices, servers, and IoT equipment. It also serves as the first point of network security enforcement.

This model improves scalability, performance, manageability, and fault isolation across large enterprise networks.

2. VLANs and Segmentation

Virtual LANs (VLANs) logically divide the network into isolated segments, each with a specific role or department. This enhances both security and traffic efficiency by limiting broadcast domains. The design includes:

- VLAN 10 – Retail Stores 1
- VLAN 20 – Retail Stores 2
- VLAN 30 –Retail Stores 3
- VLAN 40 – Food Court (Customer Wi-Fi)
- VLAN 50 – CCTV Surveillance
- VLAN 60 – Administrative Management
- VLAN 70 – Internal Server Services
- VLAN 80 – IoT Systems (AC units, alarms, sensors)

This segmentation ensures that devices only communicate where necessary and improves control over traffic flows.

3. Routing and Redundancy

Inter-VLAN routing is handled by Switched Virtual Interfaces (SVIs) configured on the Layer 3 distribution switches. These switches, along with the core router, run OSPF (Open Shortest Path First), a dynamic routing protocol that maintains updated route information across the network.

To ensure availability and fault tolerance, Hot Standby Router Protocol (HSRP) is configured between the distribution switches, allowing one to act as a backup gateway in case the other fails. Additionally, a passive OSPF interface is configured between them to prevent unwanted neighbor relationships while maintaining route awareness.

4. IP Addressing and DHCP Configuration

The Layer 3 distribution switches function as DHCP servers for VLANs 10 through 60, dynamically assigning IP addresses to user devices, printers, and other endpoints within those network segments. This approach provides efficient address management, reduces dependency on a centralized DHCP server, and enhances network resilience and scalability.

Separately, a dedicated DHCP server is deployed in VLAN 80 to manage IP addressing for IoT devices such as fire alarms, sprinklers, air conditioning units, and detectors. This configuration ensures these devices receive valid IP addresses automatically, simplifying deployment and enabling seamless communication and centralized management as the number of IoT systems grows.

IP Addressing Scheme (VLAN):

- VLAN 10: 192.168.10.0/24
- VLAN 20: 192.168.20.0/24
- VLAN 30: 192.168.30.0/24
- VLAN 40: 192.168.40.0/24
- VLAN 50: 192.168.50.0/24
- VLAN 60: 192.168.60.0/24
- VLAN 70: 192.168.70.0/24
- VLAN 80: 192.168.80.0/24

IP Addressing scheme between distribution switch 1 and core router : 10.10.10.0 /30

IP Addressing scheme between distribution switch 2 and core router : 10.10.10.4 /30

IP Addressing scheme between distribution core router and isp router : 100.1.1.0 /30

IP Addressing scheme of external server : 172.16.1.0 /24

5. Access Control and Security

Security is enforced using Access Control Lists (ACLs) applied to VLAN interfaces to manage inter-VLAN traffic. VLANs assigned to the retail store, food court, CCTV security, IoT devices, and administrative systems are fully isolated from one another, preventing direct communication between them. Only the administrative VLAN is permitted to initiate traffic to the other VLANs, enabling centralized monitoring, updates, and control while maintaining strict segmentation.

Server access is divided between internal and external services. Internal servers, hosting critical applications and resources, are accessible exclusively from the administrative VLAN to protect sensitive data and administrative functions.

Port security is enabled on access layer switches to limit the number of devices per port and bind them to specific MAC addresses, preventing unauthorized device connections. Remote management is secured via SSH, ensuring encrypted administrative access and eliminating the use of insecure protocols like Telnet.

6. Spanning Tree Protocol (STP) and PortFast

To prevent switching loops in a topology with redundant links between access and distribution switches, Spanning Tree Protocol (STP) is enabled by default on all Cisco switches. It ensures a loop-free Layer 2 topology by dynamically blocking redundant paths. While no manual tuning (e.g. root bridge priority) was configured, STP ensures reliable and stable switching operations.

To improve convergence time for end devices, PortFast and BPDU Guard was enabled on all access ports. This allows those ports to transition directly to the forwarding state, bypassing the usual listening and learning stages of STP. This reduces connection delays for devices like PCs and IoT equipment without compromising loop protection, as PortFast is only applied on non-bridging ports. BPDU Guard shuts down the port if it receives a BPDU (Bridge Protocol Data Unit), preventing loops.

7. Wireless Connectivity

Two wireless access points (APs) are included in the design:

- VLAN 40 (Food Court): Provides internet access to customers using mobile devices.
- VLAN 80 (IoT): Enables wireless connectivity for devices such as air conditioners, alarms, and smart sensors.

Each AP is logically separated in its VLAN, preventing cross-traffic between public and internal systems. This ensures better control, performance, and security.

8. Server Integration

A dedicated access switch hosts in VLAN 70, which includes HTTP, FTP, and TFTP servers for internal use. These servers are only accessible from authorized VLANs, primarily the administrative and internal service zones. The isolation of server services enhances security and performance by minimizing unnecessary exposure.

9. Monitoring and Logging

A centralized syslog server positioned beyond the core router collects logs from switches and routers. This enables real-time monitoring, event auditing, and rapid troubleshooting. Logging is crucial for tracking unauthorized access, device health, and network activity over time.

10. External Services and Internet Simulation

To simulate internet connectivity, a loopback interface (IP: 8.8.8.8) was configured on the ISP router, providing a virtual target for testing NAT and routing. Additionally, a separate switch connected to the ISP router hosts external services such as HTTP, FTP, and Syslog servers. These simulate typical cloud or third-party services accessed by mail systems for updates, logging, and file transfer.

THEORY

Routing

Routing is a method used to find the best path between a source and a destination. Routers use routing tables to determine how to forward packets.

Administrative Distance (AD)

Administrative Distance is the default value assigned to routing protocols. It is used to determine which routing source to trust when a router has multiple routes to the same destination. The protocol with the lowest AD is preferred.

Routing Protocol	Administrative Distance
Static Routing	1
OSPF	110

Static Routing

- Static routes are manually configured by a network administrator.
- Suitable for small or stable networks.
- More secure, with minimal CPU usage.
- Does not adapt to link failures unless manually updated.

Command:

```
Router(config)#ip route [destination-network] [subnet-mask] [next-hop-IP]
```

Default Routing

Used when a router does not have a specific route to a destination. A default route forwards all unknown traffic. Commonly used in edge or stub networks.

Command:

```
Router(config)#ip route 0.0.0.0 0.0.0.0 [next-hop-IP]
```

Classful vs Classless Routing

- **Classful Protocols:** Do not send subnet mask info (e.g., RIPv1).
- **Classless Protocols:** Send subnet mask info. OSPF is classless.

DYNAMIC ROUTING

Advantages of OSPF Compared to Other Routing Protocols

OSPF was chosen for its fast convergence, scalability, and support for hierarchical network design. Unlike RIP, OSPF uses a link-state algorithm, providing more efficient and accurate routing. It supports features like route summarization, ECMP, and authentication, enhancing performance and security. As an open standard, it ensures compatibility across different vendors, making it more versatile than proprietary protocols like EIGRP.

OSPF (Open Shortest Path First)

- A link-state routing protocol.
- Uses **cost** as its metric (based on bandwidth).
- Sends routing updates using Link-State Advertisements (LSAs).
- Builds a complete topology of the network and calculates shortest paths using Dijkstra's algorithm.
- Forms neighbor relationships with routers in the same area.
- Supports hierarchical design using areas (e.g., Area 0 is the backbone).
- Classless: Includes subnet mask in updates.
- Administrative Distance: 110

OSPF Configuration Example

```
Router(config)#router ospf [process-id]
```

```
Router(config-router)#network [network-address] [wildcard-mask] area [area-id]
```

All routers in the same OSPF area must have matching area IDs.

Spanning Tree Protocol (STP)

Spanning Tree Protocol is a Layer 2 protocol used in switched networks to prevent loops. It creates a loop-free logical topology by identifying redundant links and placing them into a blocking state while keeping the best path active for data transmission.

In Ethernet networks, having multiple active paths can cause switching loops, leading to broadcast storms and MAC table instability. STP eliminates these issues by blocking redundant paths and ensuring only one active path exists between two network devices. PVST+ (Per VLAN STP): Cisco proprietary; runs separate STP instance per VLAN. In cisco switch STP is turned on by default.

STP Timers (Default)

- Hello Time: 2 seconds
- Forward Delay: 15 seconds
- Max Age: 20 seconds

STP Verification Commands

```
show spanning-tree
```

```
show spanning-tree vlan [vlan-id]
```

```
show spanning-tree summary
```

Virtual Local Area Network (VLAN)

Virtual Local Area Network (VLAN) is a logical grouping of devices within a network, regardless of their physical location. VLANs allow network administrators to segment a single physical network into multiple isolated broadcast domains, improving network performance, security, and manageability.

VLANs are used to separate large networks into smaller, more manageable segments. Devices in the same VLAN can communicate directly, while communication between VLANs requires a Layer 3 device (router or Layer 3 switch).

Characteristics of VLAN

- Logical Segmentation: Devices are grouped based on function, department, or application—not physical location.
- Broadcast Control: Each VLAN is its own broadcast domain, reducing unnecessary traffic.
- Improved Security: Devices in different VLANs cannot communicate without routing, providing isolation.
- Efficient Management: VLANs simplify configuration and network changes.

VLAN Configuration example

Creating a VLAN:

```
Switch(config)#vlan [vlan-id]
```

```
Switch(config-vlan)#name [vlan-name]
```

Assigning a VLAN to a Port:

```
Switch(config)#interface [interface-id]
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan [vlan-id]
```

Verifying VLANs:

```
Switch#show vlan brief
```

Trunk Configuration:

```
Switch(config)#interface [interface-id]
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#switchport trunk allowed vlan [vlan-list]
```

Inter-VLAN routing

Since VLANs are separate broadcast domains, communication between them requires a Layer 3 device.

Options for Inter-VLAN Routing:

- Router-on-a-stick: A single router interface configured with sub-interfaces for each VLAN.
- Layer 3 Switch: Performs routing directly between VLANs using SVI (Switched Virtual Interfaces).

Switched Virtual Interface (SVI)

A Switched Virtual Interface (SVI) is a virtual Layer 3 interface configured on a switch to enable routing between VLANs. It acts as the default gateway for devices within a VLAN, allowing inter-VLAN communication without the need for an external router.

Provides Layer 3 connectivity between VLANs on a Layer 3 switch and provides Layer 3 connectivity between VLANs on a Layer 3 switch. This eliminates the need for a separate router in inter-VLAN routing scenarios (replaces router-on-a-stick).

SVI Configuration example:

Configure the SVI

```
Switch(config)#interface vlan [vlan-id]
Switch(config-if)#ip address [ip-address] [subnet-mask]
Switch(config-if)#no shutdown
```

Enable IP Routing

```
Switch(config)#ip routing
```

Hot standby Router Protocol (HSRP)

Hot Standby Router Protocol (HSRP) is a Cisco proprietary redundancy protocol used in networks to ensure high availability of default gateways. It provides gateway redundancy by allowing multiple routers to work together to present the illusion of a single virtual router (default gateway) to the hosts on a LAN.

HSRP provides fault tolerance for IP traffic by ensuring that user traffic immediately and transparently recovers from a failure of the first hop router. If the active router fails, another router in the HSRP group automatically takes over as the new active router without requiring any changes to end devices.

HSRP Operation

- Routers in the same HSRP group send hello messages to each other at regular intervals.
- The router with the highest priority becomes the active router (default priority is 100).
- If the active router fails, the standby router takes over using the same virtual IP and MAC.
- Traffic continues to flow through the network without interruption.

HSRP Configuration Example

```
Switch(config)#interface [interface-id]
Switch(config-if)#standby [group-number] ip [virtual-ip-address]
Switch(config-if)#standby [group-number] priority [value]
Switch(config-if)#standby [group-number] preempt
```

Dynamic Host Configuration Protocol (DHCP)

DHCP (Dynamic Host Configuration Protocol) is a network protocol used to automatically assign IP addresses and other network configuration settings (such as subnet mask, default gateway, and DNS server) to client devices on a network. This eliminates the need for manual IP configuration on each device.

Example Configuration of a DHCP server on a router:

```
// Exclude specific IPs from being assigned
```

```
Router(config)#ip dhcp excluded-address [start-ip] [end-ip]
Router(config)#ip dhcp pool [pool-name]
Router(dhcp-config)#network [network-address] [subnet-mask]
Router(dhcp-config)#default-router [default-gateway]
```

Access control list (Acl)

- A set of rules which will allow or deny the specific traffic moving through the route.
- It is a Layer 3 security which controls the flow of traffic from one router to another.
- It is also known as Packet filtering firewall.

Types of Acl:

1. Standard (Numbered, Name)
2. Extended (Numbered, Name)

Standard ACL: (Numbered)

- Acl number range 1-99
- We allow or deny a network, host and subnet.
- All services are blocked.
- Filtering is done based on only source IP address.
- Implemented closest to the destination.

Standard Acl Configuration

```
Router(config)# access-list(access-list number){permit/deny} host (IP address)
Router(config)# access-list (access-list number) permit any
```

Applying Acl to an interface:

```
Router(config)#interface interface name
Router(config-if)# ip access-group(access-list number {in/out})
Router(config-if)exit
```

Extended Acl: (Numbered)

- Acl number range is 100-199
- We can allow or deny a network, host, subnet and services.
- Selective services can be blocked.
- Filtering is done based on source IP , destination IP, protocol port number address.
- Implemented closet to the source.

Extended Acl configuration:

```
Router(config)#access-list <access-list num> {permit/deny} <protocol> host <source ip> host  
<destination ip> eq<port no>
```

```
Router(config)#access-list <access-list num> permit ip any any
```

Standard ACL: (Named)

Acl with name are called Named Acl's

Configuration:

```
Router(config t)#ip access-list standard "access-list name"
```

```
Router(config t- std-nacl)#deny host <address>
```

```
Router(config t-std-nacl)#permit any
```

Extended Acl: (Named)

```
Router(config t)# ip access-list extended 'access-list name'
```

```
Router(config t -ext-nacl)#deny 'service' host 'address'
```

```
Router(config t-ext-nacl)#permit icmp any any
```

Network Time Protocol (NTP)

NTP (Network Time Protocol) is used to synchronize the system clock of networking devices (routers, switches, etc.) to a centralized time source, typically a server or an external atomic clock. Accurate timekeeping is essential for network management, log accuracy, and security protocols.

NTP Configuration (Client Mode)

```
Router(config)#ntp server [ntp-server-ip]
```

Verify NTP:

```
show ntp status
```

```
show ntp associations
```

System Logging (Syslog)

Syslog is a standard protocol used to collect log messages from network devices and send them to a central Syslog server. It is used for network monitoring, troubleshooting, and auditing.

Syslog Message Severity Levels

Level	Severity	Description
0	Emergency	System is unusable
1	Alert	Immediate action needed
2	Critical	Critical condition
3	Error	Error condition
4	Warning	Warning condition
5	Notification	Normal but significant event
6	Informational	Informational messages
7	Debugging	Debug-level messages

Syslog Configuration

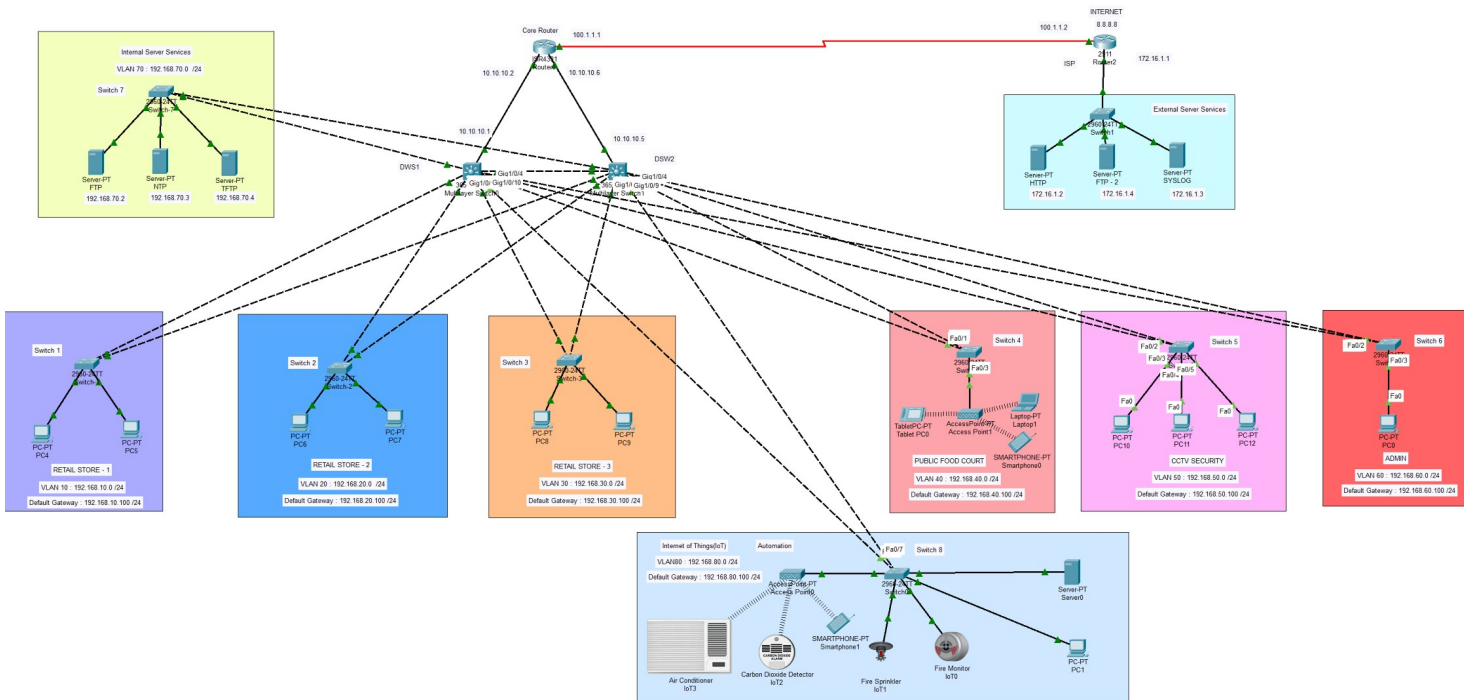
```
Router(config)#logging host [syslog-server-ip]
```

```
Router(config)#logging trap [severity-level]
```

```
Router(config)#service timestamps log datetime msec
```

```
Router(config)#logging on
```

TOPOLOGY



CONFIGURATIONS

1. IP Addressing and Interface Configuration

ISP Router

```
Router>enable
Router#configure terminal
Router(config)#hostname ISP-Router
ISP-Router(config)#exit
ISP-Router(config)# interface se0/1/0
ISP-Router(config-if)#ip add 100.1.1.2 255.255.255.252
ISP-Router(config-if)#no shutdown
ISP-Router(config-if)#exit
ISP-Router(config)# interface gig0/0
ISP-Router(config-if)#ip address 172.16.1.1 255.255.255.0
```

```
ISP-Router(config-if)#no shutdown
ISP-Router(config)# interface loopback 1
ISP-Router(config-if)#ip add 8.8.8.8 255.255.255.255
ISP-Router(config-if)#exit
```

Core Layer (Core-Router)

```
Router>enable
Router#configure terminal
Router(config)#hostname Core-Router
Core-Router(config)#interface gig0/0/0
Core-Router(config-if)#ip address 10.10.10.2 255.255.255.252
Core-Router(config-if)#no shutdown
Core-Router(config-if)#exit
Core-Router(config-if)#interface gig0/0/1
Core-Router(config-if)#ip address 10.10.10.6 255.255.255.252
Core-Router(config-if)#no shutdown
Core-Router(config-if)#exit
Core-Router(config)#interface se0/1/0
Core-Router(config-if)#ip add 100.1.1.1 255.255.255.252
Core-Router(config-if)#no shutdown
Core-Router(config-if)#exit
Core-Router(config)#interface loopback 0
Core-Router(config-if)#ip address 10.10.10.9 255.255.255.252
Core-Router(config-if)#no shutdown
Core-Router(config-if)#exit
```

SSH Configuration

Core-Router(config)#username admin privilege 15 secret 123456789

Core-Router(config)#ip domain-name mall.com

Core-Router(config)#crypto key generate rsa

How many bits in the modulus [512]: 1024

Core-Router(config)#ip ssh version 2

Core-Router(config)#line vty 0 2

Core-Router(config-line)#transport input ssh

Core-Router(config-line)#login local

Core-Router(config-line)#exit

Distribution Layer (L3 switch)

Distribution layer switch 1

Switch>enable

Switch#configure terminal

Switch(config)#hostname DSW1

DSW1(config)# interface gig1/0/7

DSW1(config-if)#no switchport

DSW1(config-if)#ip address 10.10.10.1 255.255.255.252

DSW1(config-if)#no shutdown

DSW1(config-if)#exit

SSH Configuration

DSW1(config)#username admin privilege 15 secret 123456789

DSW1(config)#ip domain-name mall.com

DSW1(config)#crypto key generate rsa

How many bits in the modulus [512]: 1024

DSW1(config)#ip ssh version 2

DSW1(config)#line vty 0 2

DSW1(config-line)#login local

DSW1(config-line)#transport input ssh

Distribution layer switch 2

Switch>enable

Switch#configure terminal

Switch(config)#hostname DSW2

DSW2(config)# interface gig1/0/7

DSW2(config-if)#no switchport

DSW2(config-if)#ip address 10.10.10.5 255.255.255.252

DSW2(config-if)#no shutdown

DSW2(config-if)#exit

SSH Configuration

DSW2(config)#username admin privilege 15 secret 123456789

DSW2(config)#ip domain-name mall.com

DSW2(config)#crypto key generate rsa

How many bits in the modulus [512]: 1024

DSW2(config)#ip ssh version 2

DSW2(config)#line vty 0 2

DSW2(config-line)#login local

DSW2(config-line)#transport input ssh

Access Layer Switch

SSH

All access layer switches use a consistent SSH configuration template. Each switch is assigned a unique management IP address on respective vlan for remote access:

example

Switch#configure terminal

Switch(config)# interface vlan 10

Switch(config-if)#ip address 192.168.10.99 255.255.255.0

Switch(config-if)#no shutdown

Switch(config-if)#exit


```
Access-SW3(config)#ip default-gateway 192.168.10.100
Access-SW1(config)#hostname Access-SW3
Access-SW1(config)#ip domain-name mall.com
Access-SW1(config)#username admin privilege 15 secret 1234
Access-SW1(config)#crypto key generate rsa
How many bits in the modulus [512]: 1024
Access-SW1(config)#ip ssh version 2
Access-SW1(config)#line vty 0 2
Access-SW1(config-line)#login local
Access-SW1(config-line)#transport input ssh
Access-SW1(config-line)#exit
```

This configuration is replicated across all access switches (Access-SW1, Access-SW2, ..., Switch-iot) with only the hostname and IP address adjusted per device.

2. Routing Configuration

ISP Router

```
ISP-Router(config)#ip route 0.0.0.0 0.0.0.0 100.1.1.1
ISP-Router(config)#exit
```

Core Layer (Core-Router)

```
Core-Router(config)#ip route 0.0.0.0 0.0.0.0 100.1.1.2
Core-Router(config)#router ospf 1
Core-Router(config-router)#network 10.10.10.2 0.0.0.0 area 0
Core-Router(config-router)#network 10.10.10.6 0.0.0.0 area 0
Core-Router(config-router)#default-information originate
Core-Router(config-router)#exit
```

Distribution Layer (L3 switch)

Distribution layer switch 1

```
DSW1(config)#router ospf 1
DSW1(config-router)#network 10.10.10.1 0.0.0.0 area 0
DSW1(config-router)#network 192.168.10.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.20.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.30.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.40.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.50.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.60.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.70.0 0.0.0.255 area 0
DSW1(config-router)#network 192.168.80.0 0.0.0.255 area 0
DSW1(config-router)#passive-interface vlan 10
DSW1(config-router)#passive-interface vlan 20
DSW1(config-router)#passive-interface vlan 30
DSW1(config-router)#passive-interface vlan 40
DSW1(config-router)#passive-interface vlan 50
DSW1(config-router)#passive-interface vlan 60
DSW1(config-router)#passive-interface vlan 70
DSW1(config-router)#passive-interface vlan 80
DSW1(config-router)#exit
```

Distribution layer switch 2

```
DSW2(config)#router ospf 1
DSW2(config-router)#network 10.10.10.5 0.0.0.0 area 0
DSW2(config-router)#network 192.168.10.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.20.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.30.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.40.0 0.0.0.255 area 0
```

```
DSW2(config-router)#network 192.168.50.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.60.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.70.0 0.0.0.255 area 0
DSW2(config-router)#network 192.168.80.0 0.0.0.255 area 0
DSW2(config-router)#passive-interface vlan 10
DSW2(config-router)#passive-interface vlan 20
DSW2(config-router)#passive-interface vlan 30
DSW2(config-router)#passive-interface vlan 40
DSW2(config-router)#passive-interface vlan 50
DSW2(config-router)#passive-interface vlan 60
DSW2(config-router)#passive-interface vlan 70
DSW2(config-router)#passive-interface vlan 80
DSW2(config-router)#exit
```

3. VLAN and Inter-VLAN Configuration

Distribution layer switch 1

VLAN

```
DSW1>en
DSW1#conf t
DSW1(config)#vlan 10
DSW1(config-vlan)#name reatil-s1
DSW1(config-vlan)#exit
DSW1(config)#vlan 20
DSW1(config-vlan)#name retail-s2
DSW1(config-vlan)#exit
DSW1(config)#vlan 30
DSW1(config-vlan)#name retail-s3
DSW1(config-vlan)#exit
DSW1(config)#vlan 40
```

```
DSW1(config-vlan)#name public
```

```
DSW1(config-vlan)#exit
```

```
DSW1(config)#vlan 50
```

```
DSW1(config-vlan)#name security
```

```
DSW1(config-vlan)#exit
```

```
DSW1(config)#vlan 60
```

```
DSW1(config-vlan)#name admin
```

```
DSW1(config-vlan)#exit
```

```
DSW1(config)#vlan 70
```

```
DSW1(config-vlan)#name server
```

```
DSW1(config-vlan)#exit
```

```
DSW1(config)#vlan 80
```

```
DSW1(config-vlan)#name iot
```

```
DSW1(config-vlan)#exit
```

```
DSW1(config)# interface range gig1/0/1-6
```

```
DSW1(config-if-range)#switchport mode trunk
```

```
DSW1(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50,60,70,80
```

```
DSW1(config-if-range)#exit
```

```
DSW1(config)# interface range gig1/0/8-10
```

```
DSW1(config-if-range)#switchport mode trunk
```

```
DSW1(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50,60,70,80
```

```
DSW1(config-if-range)#exit
```

Inter-VLAN

```
DSW1(config)# interface vlan 10
```

```
DSW1(config-if)#ip add 192.168.10.1 255.255.255.0
```

```
DSW1(config-if)#ip helper-address 192.168.10.1
```

```
DSW1(config-if)#exit
```

```
DSW1(config)# interface vlan 20
```

```
DSW1(config-if)#ip add 192.168.20.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.20.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 30
DSW1(config-if)#ip add 192.168.30.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.30.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 40
DSW1(config-if)#ip add 192.168.40.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.40.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 50
DSW1(config-if)#ip add 192.168.50.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.50.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 60
DSW1(config-if)#ip add 192.168.60.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.60.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 70
DSW1(config-if)#ip add 192.168.70.1 255.255.255.0
DSW1(config-if)#ip helper-address 192.168.80.1
DSW1(config-if)#exit
DSW1(config)# interface vlan 80
DSW1(config-if)#ip add 192.168.80.1 255.255.255.0
DSW1(config-if)#exit
```

Distribution layer switch 2

VLAN

DSW2>en

DSW2#configure terminal

DSW2(config)#vlan 10

DSW2(config-vlan)#name reatil-s1

DSW2(config-vlan)#exit

DSW2(config)#vlan 20

DSW2(config-vlan)#name retail-s2

DSW2(config-vlan)#exit

DSW2(config)#vlan 30

DSW2(config-vlan)#name retail-s3

DSW2(config-vlan)#exit

DSW2(config)#vlan 40

DSW2(config-vlan)#name public

DSW2(config-vlan)#exit

DSW2(config)#vlan 50

DSW2(config-vlan)#name security

DSW2(config-vlan)#exit

DSW2(config)#vlan 60

DSW2(config-vlan)#name admin

DSW2(config-vlan)#exit

DSW2(config)#vlan 70

DSW2(config-vlan)#name server

DSW2(config-vlan)#exit

DSW2(config)#vlan 80

DSW2(config-vlan)#name iot

DSW2(config-vlan)#exit

```
DSW2(config)# interface range gig1/0/1-6
DSW2(config-if-range)#switchport mode trunk
DSW2(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50,60,70,80
DSW2(config-if-range)#exit
DSW2(config)# interface range gig1/0/8-10
DSW2(config-if-range)#switchport mode trunk
DSW2(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50,60,70,80
DSW2(config-if-range)#exit
```

Inter-VLAN

```
DSW2(config)# interface vlan 10
DSW2(config-if)#ip add 192.168.10.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.10.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 20
DSW2(config-if)#ip add 192.168.20.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.20.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 30
DSW2(config-if)#ip add 192.168.30.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.30.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 40
DSW2(config-if)#ip add 192.168.40.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.40.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 50
DSW2(config-if)#ip add 192.168.50.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.50.2
DSW2(config-if)#exit
```

```
DSW2(config)# interface vlan 60
DSW2(config-if)#ip add 192.168.60.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.60.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 70
DSW2(config-if)#ip add 192.168.70.2 255.255.255.0
DSW2(config-if)#ip helper-address 192.168.80.2
DSW2(config-if)#exit
DSW2(config)# interface vlan 80
DSW2(config-if)#ip add 192.168.80.2 255.255.255.0
DSW2(config-if)#exit
```

Access layer

VLAN

Switch 1

```
Access-SW1>en
Access-SW1#configure terminal
Access-SW1(config)#vlan 10
Access-SW1(config-vlan)#name retail-s1
Access-SW1(config-vlan)#exit
Access-SW1(config)# interface range fa0/3-4
Access-SW1(config-if-range)#switchport mode access
Access-SW1(config-if-range)#switchport access vlan 10
Access-SW1(config-if-range)#exit
Access-SW1(config)# interface range fa0/1-2
Access-SW1(config-if-range)#switchport mode trunk
Access-SW1(config-if-range)#exit
```


switch2

```
Access-SW2>en
Access-SW2#configure terminal
Access-SW2(config)#vlan 20
Access-SW2(config-vlan)#name retail-s2
Access-SW2(config-vlan)#exit
Access-SW2(config)# interface range fa0/3-4
Access-SW2(config-if-range)#switchport mode access
Access-SW2(config-if-range)#switchport access vlan 20
Access-SW2(config-if-range)#exit
Access-SW2(config)# interface range fa0/1-2
Access-SW2(config-if-range)#switchport mode trunk
Access-SW2(config-if-range)#exit
```

switch3

```
Access-SW3>en
Access-SW3#configure terminal
Access-SW3(config)#vlan 30
Access-SW3(config-vlan)#name retail-s3
Access-SW3(config-vlan)#exit
Access-SW3(config)# interface range fa0/3-4
Access-SW3(config-if-range)#switchport mode access
Access-SW3(config-if-range)#switchport access vlan 30
Access-SW3(config-if-range)#exit
Access-SW3(config)# interface range fa0/1-2
Access-SW3(config-if-range)#switchport mode trunk
Access-SW3(config-if-range)#exit
```

switch 4

```
Access-PUBLIC>en
Access-PUBLIC#configure terminal
Access-PUBLIC(config)#vlan 40
Access-PUBLIC(config-vlan)#name public
Access-PUBLIC(config-vlan)#exit
Access-PUBLIC(config)# interface fa0/3
Access-PUBLIC(config-if-range)#switchport mode access
Access-PUBLIC(config-if-range)#switchport access vlan 40
Access-PUBLIC(config-if-range)#exit
Access-PUBLIC(config)# interface range fa0/1-2
Access-PUBLIC(config-if-range)#switchport mode trunk
Access-PUBLIC(config-if-range)#exit
```

switch 5

```
Access-Security>en
Access-Security#configure terminal
Access-Security(config)#vlan 50
Access-Security(config-vlan)#name security
Access-Security(config-vlan)#exit
Access-Security(config)# interface range fa0/3-5
Access-Security(config-if-range)#switchport mode access
Access-Security(config-if-range)#switchport access vlan 50
Access-Security(config-if-range)#exit
Access-Security(config)# interface range fa0/1-2
Access-Security(config-if-range)#switchport mode trunk
Access-Security(config-if-range)#exit
```

switch 6

```
Access-Admin>en
Access-Admin#configure terminal
Access-Admin(config)#vlan 60
Access-Admin(config-vlan)#name admin
Access-Admin(config-vlan)#exit
Access-Admin(config)# interface fa0/3
Access-Admin(config-if-range)#switchport mode access
Access-Admin(config-if-range)#switchport access vlan 60
Access-Admin(config-if-range)#exit
Access-Admin(config)# interface range fa0/1-2
Access-Admin(config-if-range)#switchport mode trunk
Access-Admin(config-if-range)#exit
```

switch 7

```
Server-1#configure terminal
Server-1(config)#vlan 70
Server-1(config-vlan)#name server
Server-1(config-vlan)#exit
Server-1(config)# interface range fa0/1-2
Server-1(config-if-range)#switchport mode access
Server-1(config-if-range)#switchport access vlan 70
Server-1(config-if-range)#exit
Server-1(config)# interface fa0/4
Server-1(config-if-range)#switchport mode access
Server-1(config-if-range)#switchport access vlan 70
Server-1(config-if-range)#exit
Server-1(config)# interface fa0/3
Server-1(config-if-range)#switchport mode trunk
```

```
Server-1(config-if-range)#exit
Server-1(config)# interface fa0/5
Server-1(config-if-range)#switchport mode trunk
Server-1(config-if-range)#exit
switch 8
Switch-iot#conf t
Switch-iot(config)#vlan 80
Switch-iot(config-vlan)#name iot
Switch-iot(config-vlan)#exit
Switch-iot(config)# interface range fa0/1-5
Switch-iot(config-if-range)#switchport mode access
Switch-iot(config-if-range)#switchport access vlan 80
Switch-iot(config-if-range)#exit
Switch-iot(config)# interface range fa0/6-7
Switch-iot(config-if-range)#switchport mode trunk
Switch-iot(config-if-range)#exit
```

4. Spanning Tree and Redundancy

Distribution layer switch 1

HSRP configuration

```
DSW1(config)# interface vlan 10
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.10.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 20
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.20.100
DSW1(config-if)#exit
```

```
DSW1(config)# interface vlan 30
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.30.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 40
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.40.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 50
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.50.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 60
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.60.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 70
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.70.100
DSW1(config-if)#exit
DSW1(config)# interface vlan 80
DSW1(config-if)#standby 10 priority 110
DSW1(config-if)#standby 10 preempt
DSW1(config-if)#standby 10 ip 192.168.80.100
```

DSW1(config-if)#exit

Distribution layer switch 2

HSRP configuration

DSW2(config)# interface vlan 10

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.10.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 20

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.20.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 30

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.30.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 40

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.40.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 50

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.50.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 60

DSW2(config-if)#standby 10 preempt

DSW2(config-if)#standby 10 ip 192.168.60.100

DSW2(config-if)#exit

DSW2(config)# interface vlan 70

DSW2(config-if)#standby 10 preempt

```
DSW2(config-if)#standby 10 ip 192.168.70.100
```

```
DSW2(config-if)#exit
```

```
DSW2(config)# interface vlan 80
```

```
DSW2(config-if)#standby 10 preempt
```

```
DSW2(config-if)#standby 10 ip 192.168.80.100
```

```
DSW2(config-if)#exit
```

Access Layer switch

STP configuration

switch 1

```
Access-SW1(config)#interface range fa0/3-4
```

```
Access-SW1(config-if-range)#spanning-tree portfast
```

```
Access-SW1(config-if-range)#spanning-tree bpduguard enable
```

```
Access-SW1(config-if-range)#exit
```

switch 2

```
Access-SW2(config)#interface range fa0/3-4
```

```
Access-SW2(config-if-range)#spanning-tree portfast
```

```
Access-SW2(config-if-range)#spanning-tree bpduguard enable
```

```
Access-SW2(config-if-range)#exit
```

switch 3

```
Access-SW3(config)#interface range fa0/3-4
```

```
Access-SW3(config-if-range)#spanning-tree portfast
```

```
Access-SW3(config-if-range)#spanning-tree bpduguard enable
```

```
Access-SW3(config-if-range)#exit
```

switch 4

```
Access-PUBLIC(config)#interface fa0/3
```

```
Access-PUBLIC(config-if-range)#spanning-tree portfast
```

```
Access-PUBLIC(config-if-range)#spanning-tree bpduguard enable
```

```
Access-PUBLIC(config-if-range)#exit
```

switch 5

Access-Security(config)#interface range fa0/3-5

Access-Security(config-if-range)#spanning-tree portfast

Access-Security(config-if-range)#spanning-tree bpduguard enable

Access-Security(config-if-range)#exit

switch 6

Access-Admin(config)#interface fa0/3

Access-Admin(config-if-range)#spanning-tree portfast

Access-Admin(config-if-range)#spanning-tree bpduguard enable

Access-Admin(config-if-range)#exit

switch 7

Server-1(config)#interface range fa0/3-5

Server-1(config-if-range)#spanning-tree portfast

Server-1(config-if-range)#spanning-tree bpduguard enable

Server-1(config-if-range)#exit

switch 8

Switch-iot(config)#interface range fa0/1-2

Switch-iot(config-if-range)#spanning-tree portfast

Switch-iot(config-if-range)#spanning-tree bpduguard enable

Switch-iot(config-if-range)#exit

Switch-iot(config)#interface fa0/4

Switch-iot(config-if-range)#spanning-tree portfast

Switch-iot(config-if-range)#spanning-tree bpduguard enable

Switch-iot(config-if-range)#exit

5. Security Configuration

ACL

Distribution switch 1

```
DSW1(config)#ip access-list extended BLOCK_VLAN10_TO_AllVLAN
DSW1(config-ext-nacl)#permit tcp any any established
DSW1(config-ext-nacl)#permit icmp 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 host 192.168.70.3
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip any any
DSW1(config-ext-nacl)#exit
DSW1(config)# interface vlan10
DSW1(config-if)#ip access-group BLOCK_VLAN10_TO_AllVLAN in
DSW1(config-if)#exit
```

```
DSW1(config)#ip access-list extended BLOCK_VLAN20_TO_AllVLAN
DSW1(config-ext-nacl)#permit tcp any any established
DSW1(config-ext-nacl)#permit icmp 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip 192.168.20.0 0.0.0.255 host 192.168.70.3
```

```
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.70.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.80.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#permit ip any any
```

```
DSW1(config-ext-nacl)#exit
```

```
DSW1(config)# interface vlan20
```

```
DSW1(config-if)#ip access-group BLOCK_VLAN20_TO_AllVLAN in
```

```
DSW1(config-if)#exit
```

```
DSW1(config)#ip access-list extended BLOCK_VLAN30_TO_AllVLAN
```

```
DSW1(config-ext-nacl)#permit tcp any any established
```

```
DSW1(config-ext-nacl)#permit icmp 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#permit ip 192.168.30.0 0.0.0.255 host 192.168.70.3
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.70.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.80.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#permit ip any any
```

```
DSW1(config-ext-nacl)#exit
```

```
DSW1(config)# interface vlan30
```

```
DSW1(config-if)#ip access-group BLOCK_VLAN30_TO_AllVLAN in
```

```
DSW1(config-if)#exit
```

```
DSW1(config)#ip access-list extended BLOCK_VLAN40_TO_AllVLAN
```

```
DSW1(config-ext-nacl)#permit tcp any any established
```

```
DSW1(config-ext-nacl)#permit icmp 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
```

```
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
```

```
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip 192.168.40.0 0.0.0.255 host 192.168.70.3
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip any any
DSW1(config-ext-nacl)#exit
DSW1(config)# interface vlan40
DSW1(config-if)#ip access-group BLOCK_VLAN40_TO_AllVLAN in
DSW1(config-if)#exit

DSW1(config)#ip access-list extended BLOCK_VLAN50_TO_AllVLAN
DSW1(config-ext-nacl)#permit tcp any any established
DSW1(config-ext-nacl)#permit icmp 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip 192.168.50.0 0.0.0.255 host 192.168.70.3
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip any any
DSW1(config-ext-nacl)#exit
DSW1(config)# interface vlan50
DSW1(config-if)#ip access-group BLOCK_VLAN50_TO_AllVLAN in
DSW1(config-if)#exit

DSW1(config)#ip access-list extended BLOCK_VLAN80_TO_AllVLAN
```

```
DSW1(config-ext-nacl)#permit tcp any any established
DSW1(config-ext-nacl)#permit icmp 192.168.80.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip 192.168.80.0 0.0.0.255 host 192.168.70.3
DSW1(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW1(config-ext-nacl)#permit ip any any
DSW1(config-ext-nacl)#exit
DSW1(config)#interface vlan80
DSW1(config-if)#ip access-group BLOCK_VLAN80_TO_ALLVLAN in
DSW1(config-if)#exit
```

Distribution switch 2

ACL

```
DSW2(config)#ip access-list extended BLOCK_VLAN10_TO_ALLVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.80.0 0.0.0.255
```

```
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
DSW2(config)#interface vlan10
DSW2(config-if)#ip access-group BLOCK_VLAN10_TO_AllVLAN in
DSW2(config-if)#exit
```

```
DSW2(config)#ip access-list extended BLOCK_VLAN20_TO_AllVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.20.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
DSW2(config)#interface vlan20
DSW2(config-if)#ip access-group BLOCK_VLAN20_TO_AllVLAN in
DSW2(config-if)#exit
```

```
DSW2(config)#ip access-list extended BLOCK_VLAN30_TO_AllVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
```

```
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.30.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
DSW2(config)#interface vlan30
DSW2(config-if)#ip access-group BLOCK_VLAN30_TO_AllVLAN in
DSW2(config-if)#exit

DSW2(config)#ip access-list extended BLOCK_VLAN40_TO_AllVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.40.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.40.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
DSW2(config)#interface vlan40
DSW2(config-if)#ip access-group BLOCK_VLAN40_TO_AllVLAN in
DSW2(config-if)#exit
```

```
DSW2(config)#ip access-list extended BLOCK_VLAN50_TO_AllVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.50.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.80.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
DSW2(config)#interface vlan50
DSW2(config-if)#ip access-group BLOCK_VLAN50_TO_AllVLAN in
DSW2(config-if)#exit
```

```
DSW2(config)#ip access-list extended BLOCK_VLAN80_TO_AllVLAN
DSW2(config-ext-nacl)#permit tcp any any established
DSW2(config-ext-nacl)#permit icmp 192.168.80.0 0.0.0.255 192.168.60.0 0.0.0.255 echo-reply
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.10.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.20.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.30.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.40.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.50.0 0.0.0.255
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.60.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip 192.168.80.0 0.0.0.255 host 192.168.70.3
DSW2(config-ext-nacl)#deny ip 192.168.80.0 0.0.0.255 192.168.70.0 0.0.0.255
DSW2(config-ext-nacl)#permit ip any any
DSW2(config-ext-nacl)#exit
```

```
DSW2(config)#interface vlan80
```

```
DSW2(config-if)#ip access-group BLOCK_VLAN80_TO_ALLVLAN in
```

```
DSW2(config-if)#exit
```

Access Layer Switch

Port Security

switch 1

```
Access-SW1#configure terminal
```

```
Access-SW1(config)#interface range fa0/3-4
```

```
Access-SW1(config-if-range)#switchport port-security
```

```
Access-SW1(config-if-range)#switchport port-security maximum 1
```

```
Access-SW1(config-if-range)#switchport port-security violation shutdown
```

```
Access-SW1(config-if-range)#switchport port-security mac-address sticky
```

```
Access-SW1(config-if-range)#exit
```

switch 2

```
Access-SW2#configure terminal
```

```
Access-SW2(config)#interface range fa0/3-4
```

```
Access-SW2(config-if-range)#switchport port-security
```

```
Access-SW2(config-if-range)#switchport port-security maximum 1
```

```
Access-SW2(config-if-range)#switchport port-security violation shutdown
```

```
Access-SW2(config-if-range)#switchport port-security mac-address sticky
```

```
Access-SW2(config-if-range)#exit
```

switch 3

```
Access-SW3#configure terminal
```

```
Access-SW3(config)#interface range fa0/3-4
```

```
Access-SW3(config-if-range)#switchport port-security
```

```
Access-SW3(config-if-range)#switchport port-security maximum 1
```

```
Access-SW3(config-if-range)#switchport port-security violation shutdown
```

```
Access-SW3(config-if-range)#switchport port-security mac-address sticky
```

```
Access-SW3(config-if-range)#exit
```


switch 4

Access-PUBLIC#configure terminal

Access-PUBLIC(config)#interface fa0/3

Access-PUBLIC(config-if-range)#switchport port-security

Access-PUBLIC(config-if-range)#switchport port-security maximum 1

Access-PUBLIC(config-if-range)#switchport port-security violation shutdown

Access-PUBLIC(config-if-range)#switchport port-security mac-address sticky

Access-PUBLIC(config-if-range)#exit

switch 5

Access-Security#configure terminal

Access-Security(config)#interface range fa0/3-5

Access-Security(config-if-range)#switchport port-security

Access-Security(config-if-range)#switchport port-security maximum 1

Access-Security(config-if-range)#switchport port-security violation shutdown

Access-Security(config-if-range)#switchport port-security mac-address sticky

Access-Security(config-if-range)#exit

switch 6

Access-Admin#configure terminal

Access-Admin(config)#interface fa0/3

Access-Admin(config-if-range)#switchport port-security

Access-Admin(config-if-range)#switchport port-security maximum 1

Access-Admin(config-if-range)#switchport port-security violation shutdown

Access-Admin(config-if-range)#switchport port-security mac-address sticky

Access-Admin(config-if-range)#exit

switch 7

Server-1#configure terminal

Server-1(config)#interface range fa0/3-4

Server-1(config-if-range)#switchport port-security

Server-1(config-if-range)#switchport port-security maximum 1

Server-1(config-if-range)#switchport port-security violation shutdown

```
Server-1(config-if-range)#switchport port-security mac-address sticky
Server-1(config-if-range)#exit
switch 8
Switch-iot#configure terminal
Switch-iot(config)#interface range fa0/3-4
Switch-iot(config-if-range)#switchport port-security
Switch-iot(config-if-range)#switchport port-security maximum 1
Switch-iot(config-if-range)#switchport port-security violation shutdown
Switch-iot(config-if-range)#switchport port-security mac-address sticky
Switch-iot(config-if-range)#exit
```

6. DHCP Dynamic ip addressing

Distribution layer-3 switch 1

```
DSW1(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.2
DSW1(config)#ip dhcp pool vlan10_rs1
DSW1(dhcp-config)#network 192.168.10.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.10.100
DSW1(dhcp-config)#exit
DSW1(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.2
DSW1(config)#ip dhcp pool vlan20_rs2
DSW1(dhcp-config)#network 192.168.20.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.20.100
DSW1(dhcp-config)#exit
DSW1(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.2
DSW1(config)#ip dhcp pool vlan30_rs3
DSW1(dhcp-config)#network 192.168.30.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.30.100
DSW1(dhcp-config)#exit
DSW1(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.2
```

```
DSW1(config)#ip dhcp pool vlan40_public
DSW1(dhcp-config)#network 192.168.40.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.40.100
DSW1(dhcp-config)#exit
DSW1(config)#ip dhcp excluded-address 192.168.50.1 192.168.50.2
DSW1(config)#ip dhcp pool vlan50_security
DSW1(dhcp-config)#network 192.168.50.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.50.100
DSW1(dhcp-config)#exit
DSW1(config)#ip dhcp excluded-address 192.168.60.1 192.168.60.2
DSW1(config)#ip dhcp pool vlan60_admin
DSW1(dhcp-config)#network 192.168.60.0 255.255.255.0
DSW1(dhcp-config)#default-router 192.168.60.100
DSW1(dhcp-config)#exit
```

Distribution Layer-3 Switch 2

```
DSW2(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.2
DSW2(config)#ip dhcp pool vlan10_rs1
DSW2(dhcp-config)#network 192.168.10.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.10.100
DSW2(dhcp-config)#exit
DSW2(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.2
DSW2(config)#ip dhcp pool vlan20_rs2
DSW2(dhcp-config)#network 192.168.20.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.20.100
DSW2(dhcp-config)#exit
DSW2(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.2
DSW2(config)#ip dhcp pool vlan30_rs3
DSW2(dhcp-config)#network 192.168.30.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.30.100
```

```
DSW2(dhcp-config)#exit
DSW2(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.2
DSW2(config)#ip dhcp pool vlan40_public
DSW2(dhcp-config)#network 192.168.40.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.40.100
DSW2(dhcp-config)#exit
DSW2(config)#ip dhcp excluded-address 192.168.50.1 192.168.50.2
DSW2(config)#ip dhcp pool vlan50_security
DSW2(dhcp-config)#network 192.168.50.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.50.100
DSW2(dhcp-config)#exit
DSW2(config)#ip dhcp excluded-address 192.168.60.1 192.168.60.2
DSW2(config)#ip dhcp pool vlan60_admin
DSW2(dhcp-config)#network 192.168.60.0 255.255.255.0
DSW2(dhcp-config)#default-router 192.168.60.100
DSW2(dhcp-config)#exit
```

7. Network Address Translation (NAT)

```
Core-Router(config)#access-list 10 permit 192.168.0.0 0.0.255.255
Core-Router(config)#ip nat inside source list 10 interface se0/1/0 overload
Core-Router(config)#interface se0/1/0
Core-Router(config-if)#ip nat outside
Core-Router(config-if)#exit
Core-Router(config)#interface gig0/0/0
Core-Router(config-if)#ip nat inside
Core-Router(config-if)#exit
Core-Router(config)#interface gig0/0/1
Core-Router(config-if)#ip nat inside
Core-Router(config-if)#exit
```

8. NTP and SYSLOG

NTP

```
Access-SW1(config)#ntp server 192.168.70.3
```

```
Access-SW1(config)#exit
```

SYSLOG

```
Access-SW1(config)#logging on
```

```
Access-SW1(config)#logging host 172.16.1.3
```

```
Access-SW1(config)#logging trap debugging
```

```
Access-SW1(config)#service timestamps log datetime msec
```

```
Access-SW1(config)#end
```

The same configurations mentioned above is applied to all the switches in the access layer ,distribution layer-3 switches and core router.

RESULTS

The final implementation of the shopping mall network was tested for functionality, reliability, and security across all configured components. The network successfully met its design objectives, with proper segmentation, secure access, and seamless communication between critical systems.

1. VLAN Functionality

All VLANs were correctly created and assigned across the access and distribution layers. Devices within the same VLAN could communicate without issue, and inter-VLAN communication was successfully handled by SVIs on the distribution switches.

2. DHCP and IP Allocation

Dynamic IP assignment worked as expected for all VLANs. The DHCP server configured in VLAN 80 successfully assigned IP addresses to IoT devices. Distribution switches assigned IP addresses to their respective VLANs, and no IP conflicts were detected during testing.

3. Internet Access

NAT on the core router allowed internal devices to access external servers hosted on the ISP-connected switch. Devices could successfully reach the loopback address (8.8.8.8), simulating internet connectivity. HTTP and FTP traffic to the external servers was verified using end devices in the mall network.

4. Routing and Redundancy

OSPF routing was fully functional, with all distribution switches successfully forming OSPF neighbor relationships with the core router. Passive interfaces prevented unnecessary OSPF advertisements between distribution switches. HSRP failover was validated by manually disabling the active router, with traffic shifting seamlessly to the standby router.

5. Wireless Connectivity

Wireless Access Points deployed in VLANs 40 and 80 provided stable connectivity. Devices in the food court and wireless IoT devices were able to connect and receive DHCP-assigned IP addresses, confirming AP and VLAN association.

6. Security Measures

Access Control Lists (ACLs) properly restricted inter-VLAN communication based on defined policies. Retail, food court, CCTV, and IoT VLANs were isolated from one another, while the administrative VLAN retained full access. SSH was tested and verified for secure remote switch and router access. Port security was also effective in blocking unauthorized MAC addresses.

7. Logging and Monitoring

The Syslog server successfully received logs from internal network devices, enabling centralized monitoring and event tracking. Additionally, IoT monitoring functionality was verified: both a smartphone and a computer were able to access the IoT server within VLAN 80, allowing for successful observation and control of connected IoT devices.

8. Internal and External Server Services

A set of internal servers was configured to provide FTP, TFTP, and NTP services within a dedicated server VLAN. These services are critical for file transfer, configuration backup, and time synchronization across the network. Access Control Lists (ACLs) were applied at the distribution layer to ensure that only devices within the administrative VLAN (VLAN 60) could reach these internal services. This setup maintains tight control over sensitive operations and reinforces the principle of least privilege within the network.

The external servers connected via the ISP router were successfully accessed for HTTP browsing, file transfer through FTP, and centralized logging via Syslog, confirming functional connectivity between internal clients and simulated internet services.

CONCLUSION

The implemented network demonstrates a secure, scalable, and fault-tolerant architecture suitable for a modern shopping mall. It supports diverse operational needs including retail services, IoT device control, administrative management, and secure external access.